

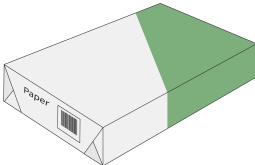
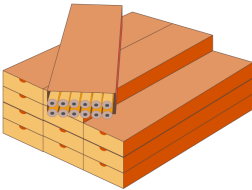
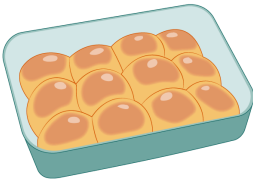
The mole and Avogadro constant

Learning outcomes

By the end of this section you should be able to:

- Explain the use of the term mole to represent the amount of elementary entities, which includes atoms, molecules and formula units.
- Calculate amounts of substance n (given in moles) to the total number of specified elementary entities (atoms, molecules, ions and electrons).

Different units are used to count groups of items: a pair means two items, a dozen contains 12 items, a gross contains 144 items and a ream contains 500 items (**Interactive 1**). A pair will always represent two items, whether it represents the number of eggs, test tubes or people. Counting a large quantity of items arranged into groups allows for the process to be manageable, which is more practical for counting such large quantities.



ream

pair

dozen

gross

☒ Check

Interactive 1. Match the Unit of Measurement for Each of These Objects.

You may have noticed above that the smaller the object, the greater the number of items that are grouped together. For instance, a pair of gloves or shoes containing two larger items compared to a ream of paper containing five hundred thin sheets of paper. Can you imagine the number of tiny atoms that might be grouped together?

The mole (mol) is the unit used to measure the amount of substance (n). One mole contains exactly the number of elementary entities given by the Avogadro constant (or N_A) (**Figure 1**). It was named after the Italian scientist Amedeo Avogadro, and its value is given in [section 2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/>) of the DP chemistry data booklet as $6.02 \times 10^{23} \text{ mol}^{-1}$. One mole is used to represent the number of elementary entities, but it could also be used to represent 6.02×10^{23} of anything.

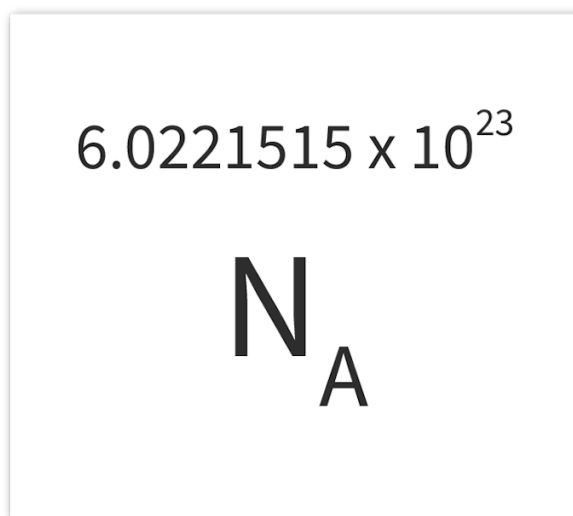

$$6.0221515 \times 10^{23}$$
$$N_A$$

Figure 1. The Avogadro constant. For calculations, this number is rounded to 6.02×10^{23} .

 More information for figure 1

The image displays the notation for Avogadro's constant. At the top, there is a number written in scientific notation as " 6.0221515×10^{23} ". Below this, centered in the image, is the letter "N" with a subscript "A" (N_A), which denotes Avogadro's number. The image is enclosed within a rectangular border, providing a neat presentation of the information.

[Generated by AI]

Do you think you now understand what a mole represents? Can you create an image in your mind of how big this number is? Watch **Video 1** to dig deeper into the concept of the mole and to put the magnitude of the Avogadro constant into perspective.

Introduction to Moles



Video 1. An introduction to moles.

Recall from [section S1.1.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>) that pure substances are classified as elements and compounds. In this section, we are going to introduce a new unit, named the mole, that represents an amount of chemical particles, known as elementary entities (see **Table 1**). Elementary entities can be atoms, molecules, ions, electrons or any other chemical particle, such as a formula unit. A formula unit represents the simplest ratio of ions in an ionic compound, for example sodium and chloride ions bond in a 1:1 ratio, making the formula NaCl. You will learn more about formula units in [section S2.1.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>).

Table 1. The number of particles in one mole of some elementary entities.

Elementary entity	Type of matter	Example	Number of particles
1 mol of atoms	element	Fe	6.02×10^{23} atoms
1 mole of molecules	molecular compound	H ₂ O	6.02×10^{23} molecules
1 mole of ions	ion	Cl [−]	6.02×10^{23} ions
1 mole of electrons	subatomic particle	e [−]	6.02×10^{23} electrons

Nature of Science

Aspect: Measurement

The Avogadro constant is a very large number. But it is a good example of a great tool that science has developed to help us measure matter at the atomic scale.

How big is a mole? (Not the animal, the other one.) - Da...



Video 2. How big is a mole?

Study skills

One mole of any substance contains 6.02×10^{23} elementary entities of that substance. This number is given in the DP chemistry data booklet so you don't need to memorise it, but you will have to be confident in using it in your calculations and conversions. Be sure that you are correctly using the exponent functions on your calculator for these calculations.

We use the Avogadro constant to calculate the number of atoms, molecules, ions or electrons. Since the unit for the Avogadro constant is mol^{-1} , this quantity allows for the units to vary, depending on the type of elementary entity being calculated (**Table 2**).

Table 2. Units of different elementary entities.

Elementary entity	Avogadro constant
atoms	$6.02 \times 10^{23} \text{ atoms mol}^{-1}$
molecules	$6.02 \times 10^{23} \text{ molecules mol}^{-1}$
ions	$6.02 \times 10^{23} \text{ ions mol}^{-1}$
electrons	$6.02 \times 10^{23} \text{ electrons mol}^{-1}$

Theory of Knowledge

The magnitude of the Avogadro constant of 6.02×10^{23} is well beyond the scale of quantities of objects in our daily lives. Even the number of stars in the Milky Way galaxy is estimated to be only 100–400 billion stars. To what extent does our ability to comprehend this magnitude limit our understanding of the quantity of elementary entities in 1 mole?

Let's do a few examples using the Avogadro constant.

Worked example 1

Calculate the number of atoms in 1.00 mole of carbon.

Solution steps	Calculations
Step 1: recognise that the Avogadro constant will be required.	The question involves finding a given number of moles so the Avogadro constant will be required.
Step 2: write out the values given in the question. The value for the Avogadro constant is in the DP chemistry data booklet.	Amount of carbon = 1.00 Avogadro constant = 6.02×10^{23}
Step 3: Identify the equation used to find the number of atoms.	Number of atoms = number of moles $\times$ Avogadro constant
Step 4: substitute the values into the formula.	Number of atoms = $1.00 \times 6.02 \times 10^{23}$
Step 5: state the answer with the appropriate number of significant figures used in rounding.	$= 6.02 \times 10^{23}$ atoms of carbon

Worked example 2

Calculate the number of moles in 1.20×10^{24} atoms of iron.

Solution steps	Calculations
Step 1: recognise that Avogadro's constant will be required.	The question involves finding a given number of carbon constant will be required.
Step 2: write out the values given in the question. The value for the Avogadro constant is in the DP chemistry data booklet.	Amount of iron = 1.20×10^2 Avogadro constant = 6.02×10^{23}
Step 3: Identify the equation used to find the number of atoms.	Number of atoms = $\frac{\text{number of moles} \times \text{Avogadro constant}}{\text{atoms per mole}}$
Step 4: make the number of moles the subject of the formula.	Number of moles = $\frac{\text{number of atoms}}{\text{Avogadro constant}}$
Step 5: substitute the values into the formula.	Moles of Fe = $\frac{1.20 \times 10^2}{6.02 \times 10^{23}}$ $= 1.9934 \times 10^{-22}$
Step 6: state the answer with the appropriate number of significant figures used in rounding.	$= 1.99$ (to 3 s.f.)

Worked example 3

Determine the number of water molecules in 2.50 moles of water.

Solution steps	Calculations
Step 1: recognise that the Avogadro constant will be required.	The question involves finding a given number of moles so the Avogadro constant will be required.
Step 2: write out the values given in the question.	Amount of water = 2.50 moles Avogadro constant = 6.02×10^{23}

Solution steps	Calculations
The value for the Avogadro constant is in the DP chemistry data booklet.	
Step 3: Identify the equation used to find the number of molecules.	Number of molecules = $n \times N_A$
Step 4: substitute the values into the formula.	Number of molecules = 2.51×10^{24}
Step 5: state the answer with the appropriate number of significant figures used in rounding.	$= 1.51 \times 10^{24}$ molecules of CH_4

Worked example 4

Calculate the number of electrons in 3.00 moles of carbon, C.

Solution steps	Calculations
Step 1: recognise that the Avogadro constant will be required.	The question involves finding a given number of moles so the Avogadro constant will be required.
Step 2: write out the values given in the question. The value for the Avogadro constant is in the DP chemistry data booklet.	Amount of carbon = 3.00 mol Avogadro constant = 6.02 × 10 ²³ mol ⁻¹ Number of electrons in 1 carbon atom = 6
Step 3: Identify the equation used to find the number of electrons.	Number of electrons = 6 × 6.02 × 10 ²³ × 3.00
Step 4: substitute the values into the formula	Number of atoms = 6 × 3.00
Step 5: state the answer with the appropriate number of significant figures used in rounding.	= 1.08 × 10 ²⁵ electrons

Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Thinking skills — Combining different ideas in order to create new understandings
- **Time required to complete activity:** 20 minutes
- **Activity type:** Pair activity

Materials required: 10 coins, blocks or other small items that can be stacked on top of one another, a ruler and pencil and paper.

1. With a partner, make a tower of the ten items and precisely measure the height with a ruler to the nearest millimetre. Calculate how high your tower would be, in kilometres, if you used 1 mole of those items. Express your answer with the correct number of significant digits. (Optional: measure the uncertainty of the ruler and propagate the uncertainty with your calculations.)
2. Using a stopwatch, measure the length of time required to do 20 jumping jacks in seconds. Calculate how many years it would take to do 1 mole of jumping jacks. Express your answer to the correct number of significant

digits. (Optional: measure the uncertainty of the stopwatch and propagate the uncertainty with your calculations)

The number of stars in the Milky Way galaxy is estimated to have at least 100 billion stars. Calculate the amount, in mol, of stars in the Milky Way.

Relative atomic mass and relative formula mass

Learning outcomes

By the end of this section you should be able to:

- Explain that all atoms have a mass that is called relative atomic mass (A_r) and is measured in comparison to the carbon-12 isotope.
- Determine the relative formula masses (M_r) from relative atomic masses (A_r) using the periodic table.

When you buy fruits and vegetables at a store or market, you are often charged based on how heavy the item is, so that you pay more money for a larger quantity. Have you ever wondered about the units of measurement that we use to determine how heavy an object is? Worldwide, there are different systems of measurement, therefore scientists use an agreed system of measurement developed by The International System of Units, abbreviated as SI.

Nature of Science

Aspect: Measurement

SI uses units of kilograms to measure the mass of an object, but where did this unit come from? SI originally used a cylinder made of an alloy of platinum and iridium that was designated as the mass of 1 kilogram and other objects were measured relative to this standard (**Figure 1**). Therefore, if an object has twice the mass of this cylinder, it is said to have a mass of 2 kilograms. If an object has half the mass of this standard, it is said to have a mass of 0.5 kilograms, which can also be expressed as 500 grams, using metric conversions.



Figure 1. The International System of Units has a reference standard of a 1 kilogram cylinder. All other objects were measured relative to this standard.

Source: "International prototype of the kilogram aka Le Grand K

(https://commons.wikimedia.org/wiki/File:International_prototype_of_the_kilogram_aka_Le_Grand_K.jpg)"

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The definition of this unit has been recently revised to be based on the Planck constant of $h = 6.63 \times 10^{-34} \text{ J s}$, which is listed in [section 2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/>) of the DP chemistry data booklet.

Chemical entities have very small masses, which would not register on a balance, so we also use a reference standard to express the masses of elements and compounds.

Recall from [section S1.2.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/atomic-structure-id-43075/>) that the particles that contribute to an atom's mass are the nuclear particles: the protons and neutrons. Protons and neutrons have almost the same mass, which is extremely small and cannot be measured directly with a balance, so all atoms are measured relative to a reference element, known as relative mass.

Chemists have chosen the carbon-12 isotope, as it contains equal numbers of protons and neutrons. The mass of the ^{12}C is assigned a value of 12, and the masses of all other atoms are compared relative to ^{12}C . For instance, an atom of the hydrogen-1 isotope, containing only one proton, is $\frac{1}{12}$ the mass of ^{12}C (**Figure 2**).

Concept

The current definition of the mole was announced in 2018 and is recommended by the IUPAC (<https://iupac.org/new-definition-mole-arrived/#:~:text=One%20mole%20contains%20exactly%206.022,is%20called%20the%20Avogadro%2>

The mole was previously defined as the amount of substance containing the same number of particles as the number of atoms contained in exactly 12 grams of the isotope carbon-12 (see [section S1.2.2](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/>)).

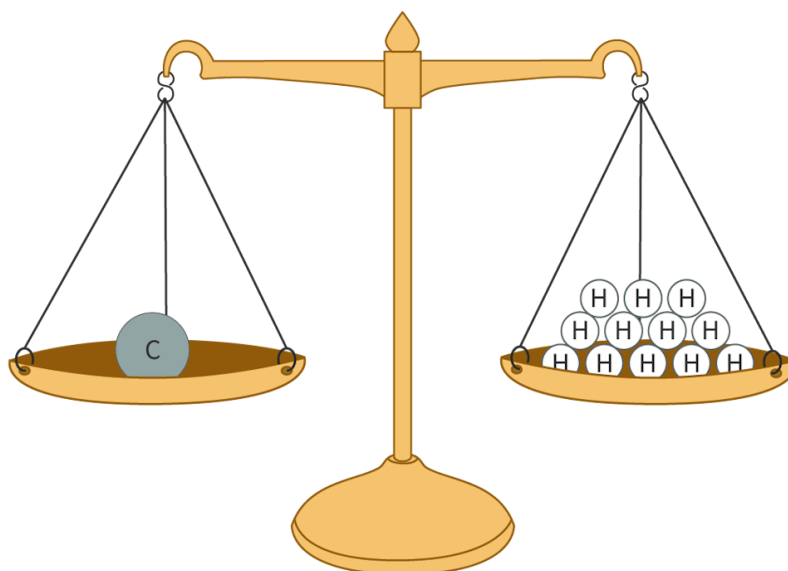


Figure 2. Hydrogen atoms are $\frac{1}{12}$ the mass of one carbon atom.

 More information for figure 2

The image depicts a balance scale. On the left side of the scale, there is a single sphere labeled with a "C," representing a carbon atom. On the right side, there are twelve smaller spheres, each labeled with an "H," representing hydrogen atoms. This visual illustrates the comparison in mass between a single carbon atom and twelve hydrogen atoms, highlighting that twelve hydrogen atoms balance out the mass of one carbon atom, indicating the relative atomic mass relationship.

[Generated by AI]

In a given sample of atoms, there will be a variety of naturally occurring isotopes. Recall that you learned how to calculate the relative atomic mass for elements containing different isotopes in [section S1.2.2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/isotopes-id-43076/>) and that these values are included in the periodic table. Since atomic masses are determined as a comparison, rather than by a measurement, these masses, known as relative atomic mass (A_r), do not have units.

Making connections

Atoms increase in mass as their position descends in the periodic table. What properties might be related to this trend? See [subtopic S3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/>) to find the main patterns used to organise the elements in the periodic table.

Try the drag and drop in **Interactive 1** to match the relative atomic masses of these elements.

1.01	12.01	32.06	55.85
Carbon C	Iron Fe	Hydrogen H	Sulfur S
☒ Check			

Interactive 1. Identify the Relative Atomic Masses of These Elements.

Relative formula mass, M_r

Recall from [section S1.1.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>) that pure substances can either be elements or compounds. Elements are represented by atoms, but compounds contain two or more atoms bonded together. You should know by now that atoms of different elements have a mass that is relative to carbon-12. But what happens when atoms combine? We need to add the masses of all the atoms in the formula to calculate the relative formula mass (M_r) for that compound.

The relative atomic mass of oxygen is 16.00, as given in the periodic table ([section 7](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/>) of the DP chemistry data booklet). Therefore, the relative formula mass of the element oxygen will be $16.00 + 16.00 = 32.00$, as we know its formula is O_2 . If the form of oxygen we are studying is ozone, then the formula is O_3 , and its relative formula mass will be $16.00 \times 3 = 48.00$ (once again, notice that neither relative atomic masses nor relative formula masses have units).

Note that the IB now uses the term relative formula mass to include both molecular compounds and ionic compounds. Previously, the term relative molecular mass was used exclusively for molecular compounds and relative formula mass for ionic compounds. In this text, the term relative formula mass is used for both types of compounds.

Practise with some worked examples.

Worked example 1

Using the relative atomic masses in the periodic table, calculate the relative formula mass (M_r) of sodium chloride, NaCl.

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question requires the relative formula mass.
Step 2: write out the relative atomic masses for each element in the compound. The values can be found in the <u>periodic table</u> (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) in the DP chemistry data booklet.	$A_r(\text{Na}): 22.99$ $A_r(\text{Cl}): 35.45$
Step 3: the M_r is the sum of the A_r of all the elements in the compound.	$M_r(\text{NaCl}) =$
Step 4: add the A_r values together.	$M_r(\text{NaCl}) =$
Step 5: state the answer to the correct number of decimal places, which in this example, is two.	$= 58.44$

Worked example 2

Using the relative atomic masses in the periodic table, calculate the relative formula mass (M_r) of carbon dioxide, CO₂:

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question relative form
Step 2: write out the relative atomic masses for each element in the compound. The values can be found in the <u>periodic table</u> (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) in the DP chemistry data booklet.	$A_r(\text{C}): 12.01$ $A_r(\text{O}): 16.00$
Step 3: the M_r is the sum of the A_r of all the elements in the compound.	$M_r(\text{CO}_2) =$
Step 4: add the A_r values together.	$M_r(\text{CO}_2) =$
Step 5: state the answer to the correct number of decimal places, which in this example, is two.	$= 44.01$

Worked example 3

Using the relative atomic masses in the periodic table, calculate the relative formula mass (M_r) of the polyatomic carbonate ion, CO_3^{2-} .

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question in formula mass o
Step 2: write out the relative atomic masses for each element in the ion. The values can be found in the <u>periodic table</u> (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) in the DP chemistry data booklet.	$A_r(\text{C}): 12.01$ $A_r(\text{O}): 16.00$
Step 3: the M_r is the sum of the A_r of all the elements in the ion.	$M_r(\text{CO}_3^{2-}) =$

Solution steps	Calculations
Step 4: add the A_r values together.	$M_r(\text{CO}_3^{2-}) =$
Step 5: state the answer to the correct number of decimal places, which in this example, is two.	$= 60.01$

Note: Recall that the mass of an atom is mainly concentrated in its nuclear particles (protons and neutrons) so the extra electrons in carbonate will not change its mass significantly (neither it will the lack of electrons when the ion is positively charged).

Worked example 4

Using the relative atomic masses in the periodic table, calculate the relative formula mass (M_r) of calcium phosphate, $\text{Ca}_3(\text{PO}_4)_2$.

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involves finding the relative formula mass of a compound.
Step 2: write out the relative atomic masses for each element in the compound. The values can be found in the periodic table (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) in the DP chemistry data booklet.	$A_r(\text{Ca}): 40.08$ $A_r(\text{P}): 30.97$ $A_r(\text{O}): 16.00$
Step 3: the M_r is the sum of the A_r of all the elements in the compound.	$M_r(\text{Ca}_3(\text{PO}_4)_2) = (3 \times A_r(\text{Ca})) +$

Solution steps	Calculations
Step 4: add the A_r values together.	$M_r(\text{Ca}_3(\text{PO}_4)_2) = (3 \times 40.08) +$
Step 5: state the answer to the correct number of decimal places, which in this example, is two.	$= 310.18$

Note: when there are parentheses around a group of atoms, you must multiply the quantity of each atom by the number on the outside of the parentheses. Therefore, there are $(1 \times 2) = 2$ atoms of P and $(4 \times 2) = 8$ atoms of oxygen.

Use the [periodic table](#) to determine the relative atomic mass or relative formula mass for the following elements and compounds.

 48.00	 40.31	 2.02	 44.01
 36.46	 18.02	 32.00	 28.02
Carbon Dioxide CO₂	Water H₂O	Hydrochloric Acid HCl	Magnesium Oxide MgO
Oxygen O₂	Nitrogen N₂	Hydrogen H₂	Ozone O₃
☒ Check			

Interactive 2. Relative Atomic Masses of Some Elements.

Practical skills

Tool 1: Experimental techniques — Applying techniques

You might notice in the laboratory that some ionic compounds have H_2O in the chemical formula and are labelled as a hydrate. For example, copper(II) chloride dihydrate has the formula $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$. These are known as hydrates because they have a high attraction for water molecules. When using these substances in calculations, you must include the molar mass of the water molecules as well. For example, has a relative formula mass of:

$$A_r(\text{Cu}): 63.55 \quad (\text{Cl}): 35.45 \quad (\text{H}): 1.01 \quad (\text{O}): 16.00$$

$$M_r: 63.55 + (2 \times 35.45) + (2 \times 18.02) = 170.49$$

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Communication skills — Presenting data appropriately
- **Time required to complete activity:** 25—30 minutes
- **Activity type:** Individual activity

1. Copy and complete the table below using the [periodic table](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/>) in the DP chemistry data booklet.

Name of element	Symbol
hydrogen	
	He
nitrogen	
	O
sodium	
	Mg

Name of element	Symbol

2. Copy and complete the table below using the periodic table (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-periodic-table-id-45109/>) in the DP chemistry data booklet.

Name of compound	Symbol
water	H ₂ O
methane	CH ₄
ammonia	NH ₃
caffeine	C ₈ H ₁₀ N ₄ O ₂
aluminium nitrate	Al(NO ₃) ₃

3. Check your answers with a classmate.

Molar mass (M)

Learning outcomes

By the end of this section you should be able to apply the concepts of the mole and Avogadro's constant, and the definition of molar mass (M), to solve problems involving the relationships between number of particles, amount of substance in moles and mass in grams.

If you have ever followed a recipe, it typically begins with a list of ingredients and the quantities you need for each. The quantities are usually measured by mass or volume and precise measuring will ensure a successful and tasty outcome. Think about the type of equipment you have in your school laboratory – what might you use to measure a mole of a substance? There are no special devices to count one mole of a substance, but there is equipment for measuring the mass and volume, just like in your kitchen (**Interactive 1**).



Rights of use

Interactive 1. Measurements.

 More information for interactive 1

An interactive photograph displays various measuring scales and tools used in the kitchen. A weighing scale is placed on a wooden board, with a piece of raw fish on it, displaying a weight of 305 units. Next to the weighing scale, there is a small cup of mayonnaise, a ladle spoon holding a scoop of ice cream, various measuring tablespoons (one containing ghee), a large cup of rice, a 1 by 3 cup of walnuts, a small cup of white kidney beans, a glass container filled milk, and a small glass cup containing sauce.

Beside the wooden board, there is a cup of peas, a conical measuring cup containing flour, an opened bottle with its cap nearby, and a glass of water.

Plus signs, representing interactive hotspots, are present on the weighing scale and the glass container filled with milk. Selecting a hotspot reveals a description for the

corresponding measuring tool.

The hotspot displays the following:

Weighing scale- "Ingredients can be measured by mass in grams or kilograms."

Glass container- "Ingredients can be measured in volume by millilitres or litres."

This interactivity helps users recognize the appropriate units and containers used for measuring different types of food items, such as liquids, solids, and semi-solids in everyday cooking.

We cannot count individual elementary entities because they are too small to see and their masses and volumes are too small for any equipment to measure, but we can measure them when they are in a large grouping of 6.02×10^{23} molecules, or one mole. Grouped together makes it possible to measure the mass and volume of a substance, just like when you are following a recipe.

In the previous sections, you learned how to group elementary entities into moles and to assign relative atomic masses to individual atoms, or relative molecular masses for compounds, based on the reference to the carbon-12 isotope. When you are in the school laboratory, chemical substances are easily and accurately measured by the mass, using an analogue or electronic balance. How can we relate the number of elementary entities, grouped and counted as moles, with the carbon-12 isotope reference and measurement on a balance?

We now have to draw a relationship between the amount, in moles, of a substance with its mass. Since each element has a different number of subatomic particles affecting its mass, the relative atomic masses are used to determine the mass of one mole of a substance, known as the molar mass, abbreviated as M .

Calculations with molar mass allow us to convert the number of particles or moles of a substance into its mass in grams. The magnitude of molar mass is the same as the relative atomic mass but has the units of grams per mole (g mol^{-1}). For instance, carbon, C, has a relative atomic mass of 12.01 and its molar mass is 12.01 g mol^{-1} .

Concept

Many times, we can infer the units of a new magnitude from its own definition. For example, if we define speed as distance travelled per unit of time, then we can easily argue the possible units to be metres per second, kilometres per hour, etc.

In this section, we have defined molar mass (M) as the mass of one mole of a substance in grams. It is obvious, then, that the units of M are grams per mole (g mol^{-1}).

Can you infer other units, like the units of density (defined as amount of matter per unit of volume)?

Note: when determining units, consider that there is more than one correct combination (metres per second, kilometres per hour, miles per day...) but only one of it is accepted by the international community — the SI unit for speed is metres per second (m s^{-1}). In your calculations, you will frequently be asked to use SI units, especially in the answers. Unit conversion is an important skill for solving problems in chemistry.

Try some worked examples to check your understanding.

Worked example 1

What is the molar mass, M , of the element beryllium, Be?

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involve an element.
Step 2: write out the relative atomic mass. The values can be found in the periodic table in the DP chemistry data booklet.	$A_r(\text{Be}): 9.01$
Step 3: state the answer with appropriate units and the number of significant figures used in rounding	The molar mass M (9.01 g mol^{-1} , because A_r , is 9.01.

Worked example 2

What is the molar mass, M , of nitrogen, N_2 ?

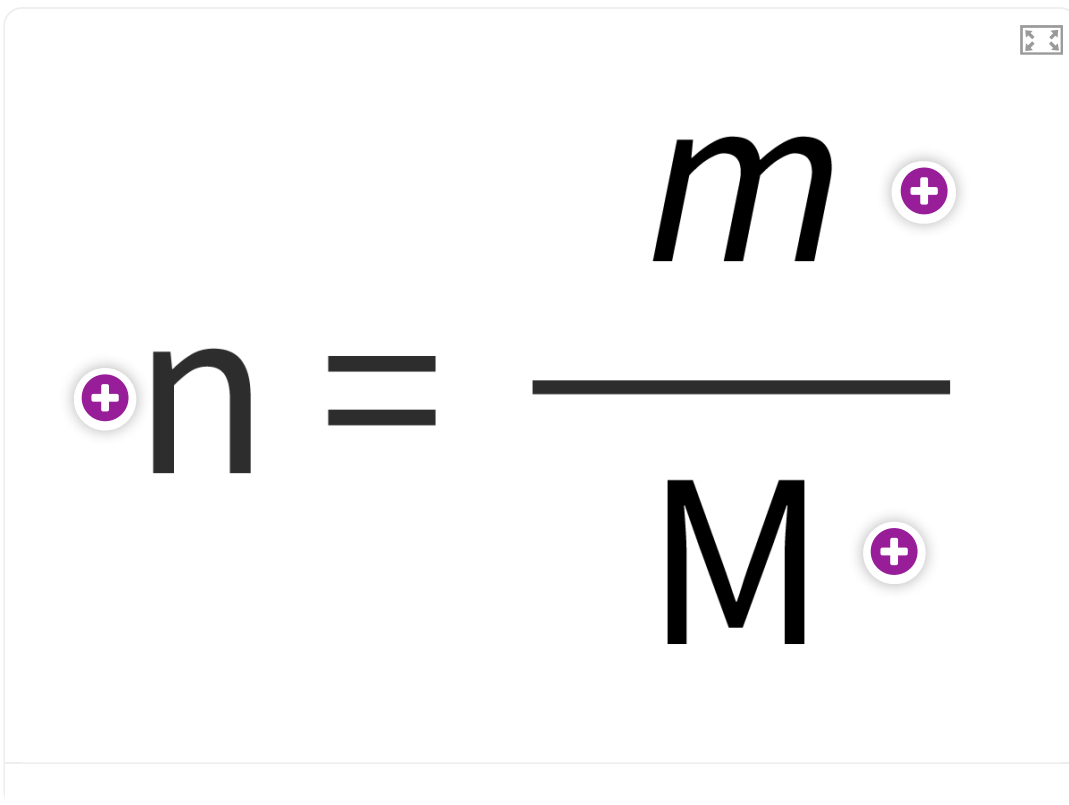
Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involves finding the r
Step 2: write out the relative atomic mass. The values can be found in the periodic table in the DP chemistry data booklet.	$A_r(\text{N}): 14.01$
Step 3: Identify the equation used to find the relative atomic mass.	relative molecular mass $M_r(\text{N}_2) =$
Step 4: substitute the values into the formula.	$M_r(\text{N}_2) = 2 \times 14.01$
Step 5: Find the relative atomic mass.	$= 28.02$
Step 6: state the answer with appropriate units and the number of significant figures used in rounding.	The molar mass $M(\text{N}_2)$ of nitrogen because its relative molecular mas

Worked example 3

What is the molar mass, M , of water, H_2O ?

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involves finding an element.
Step 2: write out the relative atomic mass. The values can be found in the periodic table in the DP chemistry data booklet.	$A_r(\text{H}): 1.01$ $A_r(\text{O}): 16.00$
Step 3: Identify the equation used to find the relative atomic mass.	$M_r(\text{H}_2\text{O}) = 2 \times A_r(\text{H}) + A_r(\text{O})$
Step 4: substitute the values into the formula.	$M_r(\text{H}_2\text{O}) = (2 \times 1.01) + 16.00$
Step 5: Find the relative atomic mass.	$= 18.02$
Step 6: state the answer with appropriate units and the number of significant figures used in rounding.	The molar mass $M(\text{H}_2\text{O})$ of water is 18.02 g mol ⁻¹ because its relative molecular mass is 18.02.

The molar mass is used when we need to determine the amount (in mol) of a substance from its mass in grams or vice versa. We can use the equation shown below in **Interactive 2**.



The diagram shows the equation $n = \frac{m}{M}$ in a large, stylized font. Each letter (n, m, and M) is accompanied by a purple circular hotspot icon with a white plus sign. The equation is presented in a clean, modern style with a white background and a thin grey border. A small icon in the top right corner of the diagram area suggests it is an interactive element.

Interactive 2. Mole Equation to Calculate the Amount from a Given Mass of a Substance.

 More information for interactive 2

The interactive represents a mole equation which is presented as follows:

$$n = \frac{m}{M}$$

The letters n, m, and M in the equation, each have a plus icon representing a hotspot. Clicking on the hotspots reveals the corresponding term for each symbol. The hotspots and the corresponding terms are as follows:

Hotspot for n: moles

Hotspot for m: mass

Hotspot for M: molar mass

This equation helps users understand the relationship between the moles, mass, and molar mass, along with their corresponding symbols.

Worked example 4

Determine the mass in grams of 3.00 moles of beryllium, Be.

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involves finding element.
Step 2: write out the relative atomic mass. The values can be found in the periodic table in the DP chemistry data booklet.	$A_r(\text{Be}): 9.01$
Step 3: Find the molar mass.	The relative atomic mass, A_r , of beryllium is 9.01. The molar mass is 9.01 g mol^{-1} . The mass of one mole of beryllium is 9.01 g.
Step 4: Identify the equation used to find the relative atomic mass.	$n = \frac{m}{M}$
Step 5: Rearrange the formula to make m the subject.	$m = n \times M$
Step 6: substitute the values into the formula.	$m = 3.00 \times 9.01$
Step 7: state the answer with appropriate units and the number of significant figures used in rounding.	$= 27.0 \text{ g}$

Worked example 5

Calculate the amount (in mol) in 15.00 grams of sodium chloride, NaCl.

Solution steps	Calculations
Step 1: recognise that a periodic table will be required.	The question involves finding the mo
Step 2: write out the relative atomic mass. The values can be found in the periodic table in the DP chemistry data booklet.	$A_r(\text{Na}): 22.99$ $A_r(\text{Cl}): 35.45$
Step 3: Find the molar mass.	$M_r(\text{NaCl}) = A_r(\text{Na}) + A_r(\text{Cl})$
Step 4: substitute the values into the formula.	$M_r = 22.99 + 35.45$ $= 58.44$
Step 5: Find the molar mass.	The relative molecular mass, M_r , of NaCl molar mass is 58.44 g mol^{-1} . That value is the mass of one mole of sodium chloride.
Step 6: Identify the equation used to find the number of moles.	$n = \frac{m}{M}$
Step 7: substitute the values into the formula.	$n = \frac{15.00}{58.44}$
Step 8: state the answer with appropriate units and the number of significant figures used in rounding	$= 0.2567 \text{ moles of NaCl}$

You might be wondering how the quantity of a substance measured in mass relates to the number of elementary entities. You cannot directly determine the number of elementary entities from its mass in grams. You must first convert the

mass to an amount in moles, then use Avogadro's constant to determine the number of atoms, molecules, ions or electrons (**Figure 1**).

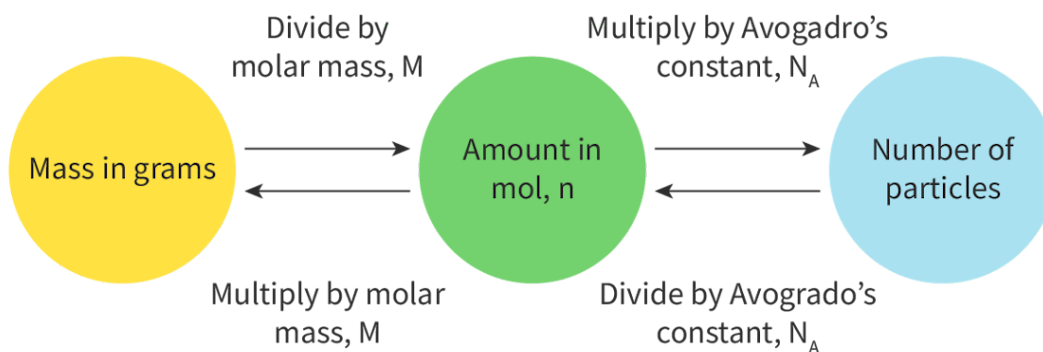


Figure 1. Relationship between the number of moles, the mass in grams and the number of particles of a substance.

 More information for figure 1

The diagram illustrates the relationship between mass in grams, the amount in moles, and the number of particles. It uses three circles labeled: "Mass in grams," "Amount in mol, n ," and "Number of particles." Arrows indicate the conversions between these quantities, with instructions provided:

1. From "Mass in grams" to "Amount in mol, n ": Divide by molar mass, M .
2. From "Amount in mol, n " to "Number of particles": Multiply by Avogadro's constant, N_A .
3. From "Number of particles" back to "Amount in mol, n ": Divide by Avogadro's constant, N_A .
4. From "Amount in mol, n " back to "Mass in grams": Multiply by molar mass, M .

The circles and text boxes visually demonstrate the flow and connections necessary for calculating between these fundamental chemistry concepts.

[Generated by AI]

Worked example 6

Given that the molar mass, M , of CO_2 is 44.01 g mol^{-1} , calculate the number of molecules in 75.0 grams of carbon dioxide, CO_2 .

Solution steps	Calculations
Step 1: Identify the equation used to find the number of moles.	$n = \frac{m}{M}$
Step 2: substitute the values into the formula.	$n = \frac{75}{44.01}$
Step 3: Find the amount (in mol).	$n = 1.70 \text{ moles of CO}_2$
Step 4: Identify the equation used to convert the number of moles into the number of molecules.	Number of molecules = number of moles $\times$ Avogadro's constant
Step 5: substitute the values into the formula.	Number of molecules = $1.70 \times 6.02 \times 10^{23}$
Step 6: state the answer with appropriate units and the number of significant figures used in rounding.	$= 1.02 \times 10^{24} \text{ molecules of CO}_2$

Worked example 7

Given that the molar mass, M , of Fe is 55.85 g mol^{-1} , calculate the mass of 1.07×10^{25} atoms of iron, Fe.

Solution steps	Calculations
Step 1: Identify the equation used to find the number of moles.	Number of atoms = number of moles $\times$ Avogadro's constant

Solution steps	Calculations
Step 2: Rearrange the formula to make number of moles the subject.	$\text{amount (in mol)} = \frac{\text{number of atoms}}{\text{Avogadro constant}}$
Step 3: substitute the values into the formula.	$\begin{aligned}\text{number of moles} &= \frac{1.07 \times 10^{25}}{6.02 \times 10^{23}} \\ &= 17.77 \text{ moles}\end{aligned}$
Step 4: Identify the equation used to convert the number of moles into the mass of the atoms.	$n = \frac{m}{M}$
Step 5: Rearrange to make m the subject.	$m = M \times n$
Step 6: substitute the values into the formula.	$m = 55.85 \times 17.77$
Step 7: state the answer with appropriate units and the number of significant figures used in rounding.	$= 992.5 \text{ g of Fe}$

Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Research skills — Using search engines and libraries effectively
- **Time required to complete activity:** 25—30 minutes
- **Activity type:** Pair activity

Throughout history and in different industries, there have been many different units of measurement. Here are some examples:

1. A unit of measurement used in the jewellery industry for gemstones is the carat (ct). A 1 carat diamond is equal to a mass of 0.200 grams. Assuming a diamond is made up only of carbon atoms, determine the number of carbon atoms in a 1 carat diamond.
2. A coal sack (sck) was a standardised unit of measurement equivalent to 112 avoirdupois pounds, which is 51 kilograms. Assuming that coal is made up only of carbon atoms, determine the amount in moles and the number of carbon atoms in 1 sack of coal.

Research another unit of measurement used for the mass of a pure substance and calculate the amount in mol and the number of elementary entities. Make a one page infographic to present your findings to your class. Learn about what an infographic (<https://youtu.be/zvmDi82xEMc>) is and design one using a program such as Canva (<https://www.canva.com/>).

Empirical formula and molecular formula

Learning outcomes

By the end of this section you should be able to:

- Interconvert the percentage composition by mass and the empirical formula of a compound.
- Determine the molecular formula of a compound from its empirical formula and molar mass.

In chemistry we use chemical formulas to represent pure substances – both elements and compounds. You may already know the chemical formulas for some substances, such as H_2O for water or CO_2 for carbon dioxide. Your school laboratory will be stocked with various elements and compounds all identified and labelled with the name and chemical formula. This is important for safety reasons to know exactly what is in each container to avoid accidentally combining the wrong substances together or in the case of a chemical spill to be sure that the substance is cleaned and disposed of properly. What if the contents of a chemical spill were unknown? How do chemists identify a substance?

There are many occasions where a substance is collected and requires identification, just like in the chemical spill example. Medical doctors perform toxicology tests on patients where there is a suspected poisoning. Processed foods and other consumer goods are tested for the presence of harmful substances to protect consumers and sporting organisations conduct doping tests on athletes to ensure that there is fair competition. All of these examples require special equipment to accurately identify an unknown substance.

Technology has evolved to develop tools like the mass spectrometer (**Figure 1**), an apparatus in which we can inject a substance (or a mix of substances) and determine its composition by mass. The percentage composition by mass of a substance gives us the information needed to determine the formula of a compound.

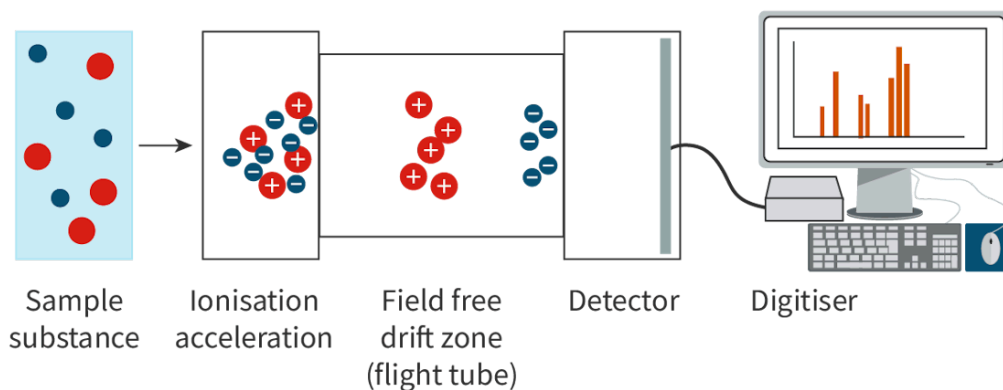


Figure 1. A mass spectrometer provides data to identify unknown substances.

 More information for figure 1

The diagram illustrates the components and process flow within a mass spectrometer. It starts with a 'Sample substance' shown on the left side, consisting of blue and red circles representing different ions.

An arrow points to the next stage labeled 'Ionisation acceleration,' where charged ions are separated based on their charge. The ions move into the 'Field free drift zone (flight tube),' depicted as a section without any imposed force, allowing ions to drift based on their initial velocity.

The ions then move to a 'Detector,' which captures their arrival and measures their mass-to-charge ratio. The detected data is then analyzed and displayed on a 'Digitiser,' shown as a computer setup on the right side of the diagram. The display on the computer screen features a graph representing the data amassed from the experimental results.

[Generated by AI]

Percentage composition by mass

All compounds are composed of elements that combine in a consistent ratio, known as the Law of Definite Proportions. For example, water molecules always have the formula H_2O – composed of exactly two hydrogen atoms and one oxygen atom. Since the ratio of atoms in the compound are always the same, we can calculate the composition of water by mass, using the molar mass for each element.

Imagine you have a business where you make bracelets using different coloured beads (**Figure 2**). Your latest product has a combination of these coloured beads, ten pink, six green and four blue. Therefore, the bracelet contains ten pink beads out of a total of twenty beads, therefore 50% of the beads are pink, 30% of the beads are green and 20% of the beads are blue. Any change to the ratio or the

number of beads would mean that the bracelet is no longer the same. It is the same with compounds – any change to the ratio or number of atoms would make it a different compound.

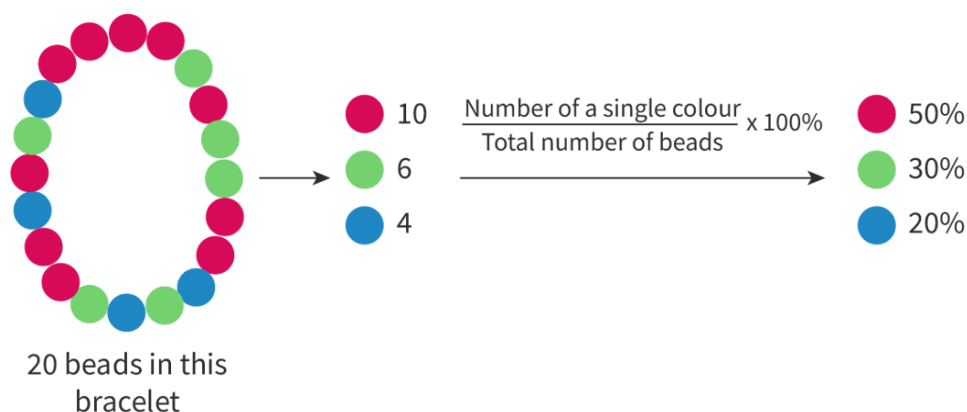


Figure 2. Beads on a bracelet are similar to the atoms in a compound.

 More information for figure 2

The image is a diagram illustrating the composition of a bracelet made up of beads in three different colors: pink, green, and blue. The bracelet consists of a series of beads in a circular arrangement. Next to the bracelet, there are corresponding numbers indicating the quantity of each color: 10 pink beads, 6 green beads, and 4 blue beads. Underneath, there is a formula shown: "Number of a single colour / Total number of beads x 100%" which explains how to compute the percentage of each color in the bracelet. The results are listed: Pink beads - 50%, Green beads - 30%, Blue beads - 20%. This highlights how the ratio and quantity of each bead contribute to their overall percentage in the bracelet composition.

[Generated by AI]

Just as you calculated the percentage of each coloured bead in the bracelet, you can do the same with the atoms in a compound. One difference, however, is that not all atoms have the same mass. We must take into consideration the mass of each element when determining its percentage in the compound. Let's do an example with water. There are twice as many atoms of hydrogen compared to oxygen per molecule, but hydrogen atoms have a much smaller mass than oxygen. Let's see how that affects the percentage composition by mass of water.

Check your understanding of percentage compositions with these worked examples.

Worked example 1

Determine the percentage composition by mass of water, H₂O.

Solution steps	Calculations
Step 1: calculate the molar mass of the compound.	$M(\text{H}_2\text{O}): (2 \times 1.01) + 16.00$
Step 2: calculate the mass that each element contributes individually to the molar mass of the compound:	$\text{H}: 2 \times 1.01 = 2.02 \text{ g mol}^{-1}$ $\text{O}: 1 \times 16.00 = 16.00 \text{ g mol}^{-1}$
Step 3: divide the mass of each element by the molar mass of the compound and multiply times 100 to get the percentage:	$\% \text{H} = \frac{2.02}{18.02} \times 100$ $= 11.21\%$ $\% \text{O} = \frac{16.00}{18.02} \times 100$ $= 88.79\%$
Step 4: state the answer with appropriate units and the number of significant figures used in rounding.	Water has a composition of 88.79% oxygen by mass.

Worked example 2

Determine the percentage composition by mass of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$.

Solution steps	Calculations
Step 1: calculate the molar mass of the compound.	$M(\text{C}_6\text{H}_{12}\text{O}_6): (6 \times 12.01) + (12 \times 1.01) + (6 \times 16.00)$

Solution steps	Calculations
Step 2: calculate the mass that each element contributes individually to the molar mass of the compound:	$\text{C: } 6 \times 12.01 = 72.06 \text{ g mol}^{-1}$ $\text{H: } 12 \times 1.01 = 12.12 \text{ g mol}^{-1}$ $\text{O: } 6 \times 16.00 = 96.00 \text{ g mol}^{-1}$
Step 3: divide the mass of each element by the molar mass of the compound and multiply times 100 to get the percentage:	$\% \text{C} = \frac{72.06}{180.18} \times 100$ $= 39.99\%$ $\% \text{H} = \frac{12.12}{180.18} \times 100$ $= 6.73\%$ $\% \text{O} = \frac{96.00}{180.18} \times 100$ $= 53.28\%$
Step 4: state the answer with appropriate units and the number of significant figures used in rounding	Glucose has a composition of 39.99% carbon 53.28% oxygen by mass.

Study skills

You can apply logic here to reason whether your answer makes sense. The percentages should always add up to 100 (or very close to 100 in the case of a rounded answer). If your answer is even 1% off, you should check to be sure that you did not make a calculation error.

You might be wondering about significant digits for these calculations. Since there are no measurements recorded, there are no guidelines for how to round your answers for this type of question. Typically, rounding your answers to two decimal places is considered sufficient for this calculation, but always check your question instructions to be sure.

Determining chemical formulas from percentage composition

Now that you have learned how to calculate the percentage by mass of a compound, you can now use this information to determine the formula for an unknown compound – essentially working in reverse.

Returning to the bracelet analogy, if a bracelet contained 40% purple beads and 60% orange beads, how many purple and orange beads are each on the bracelet? Do you have enough information to answer this question? No, because you don't know how many beads on the bracelet there are in total. The bracelet might contain two purple beads and three orange beads or four purple beads and six orange beads – or any possible ratio of two purple beads to three orange beads. To know the answer, we first have to know the total number of beads on the bracelet (**Figure 3**).

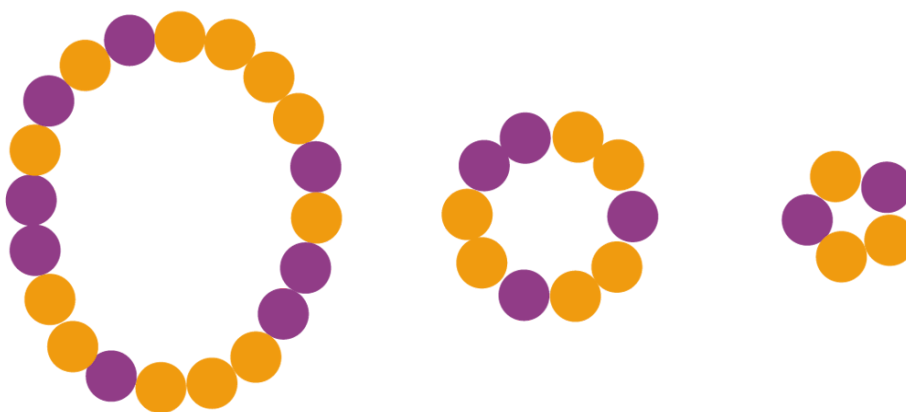


Figure 3. Each of these bracelets contain the same ratio of beads.

The same is true for compounds. There can be small compounds made up of only a few atoms or large compounds made up of hundreds or thousands of atoms – and both the small and large compounds can contain the same percentage by mass of each element. Therefore, there are two types of chemical formulas for a compound – the empirical formula and the molecular formula.

The percentage composition information will only give the simplest ratio of atoms in a compound – known as the empirical formula. Additional information is required to determine the actual number of atoms in the compound – this is the molecular formula.

Let's do an example for empirical formula first.

Worked example 3

The percentage composition by mass of sodium chloride is 39.3% sodium (Na) and 60.7% chlorine (Cl). Determine the empirical formula of sodium chloride.

Solution steps	Calculations
Step 1: make sure that percentages given add up to 100%	$39.3 + 60.7 = 100\%$
Step 2: convert the percentages to masses in grams (assuming 100 g of the compound):	Mass of Na in 100 g = Mass of Cl in 100 g = (
Step 3: convert from the mass in grams to amount (in mol) by dividing each mass by the molar mass for that element:	$n(\text{Na}) = \frac{39.3}{23.00}$ $= 1.71 \text{ mol Na}$ $n(\text{Cl}) = \frac{60.7}{35.45}$ $= 1.71 \text{ mol Cl}$
Step 4: state the answer with appropriate units and the number of significant figures used in rounding:	Since there are the same number of atoms of each element, this means the empirical formula is NaCl.

Sometimes we find lowest ratio numbers that are not whole numbers; as atoms cannot be divided, we will need to find a factor to multiply and get the lowest **whole number** ratio.

For example, if the ratio is 1.5:1, multiply both numbers by two to get 3:2; or if the ratio is 0.75:1, multiply by four to express the ratio as 3:4.

Worked example 4

A compound was collected and analysed to contain 25.94% nitrogen and the rest is oxygen. Determine the empirical formula.

Solution steps	Calculations
Step 1: first, determine the percentage of oxygen by subtracting the mass of nitrogen from 100%.	$100 - 25.94 = 74.06\% \text{ O}$
Step 2: convert the percentages to masses in grams (assuming 100 g of the compound):	Mass of N in 100 g = 25.94 Mass of O in 100 g = 74.06
Step 3: convert from the mass in grams to amount (in mol) by dividing each mass by the molar mass for that element:	$n(\text{N}) = \frac{25.94}{14.01} = 1.851 \text{ mol N}$ $n(\text{O}) = \frac{74.06}{16.00} = 4.629 \text{ mol O}$
Step 4: divide each value by the smallest amount in mol.	N: $\frac{1.851}{1.851} = 1.0$ O: $\frac{4.629}{1.851} = 2.5$
Step 5: state the answer with appropriate units and the number of significant figures used in rounding:	This gives the ratio of atom must contain whole number ratio must be multiplied by ratio of 2:5. This makes the

Returning to the bracelet analogy, the total number of beads on the bracelet, combined with the percentage of each colour, is required to know how many of each coloured bead is on the bracelet. Similarly for compounds, the total mass of the compound is required to determine the number of each atom. The total mass is determined by a mass spectrometer. To know the molecular formula, you must compare the data from the mass spectrometer with that of the empirical formula.

For some compounds, the empirical and molecular formulas are the same. For example, water is H_2O – the ratio of two hydrogen atoms to one oxygen atom is the simplest whole number ratio of atoms and it is also the actual number of atoms in the molecule. For other compounds, the molecular formula is a multiple of the empirical formula.

Consider the compound glucose. It is made of six atoms of carbon, twelve atoms of hydrogen and six atoms of oxygen in each molecule. The molecular formula is therefore $\text{C}_6\text{H}_{12}\text{O}_6$, which is different from the simplest ratio of atoms of 1C: 2H: 1O, which is the empirical formula, CH_2O .

Worked example 5

The percentage composition by mass of aluminium chloride is 20.3% aluminium (Al) and 79.7% chlorine (Cl). Determine the empirical formula of aluminium chloride. The relative formula mass of the compound, M_r , is 267. Use this value to calculate the molecular formula of the compound.

Solution steps	Calculations
Step 1: the first step is to make sure that percentages given add up to 100%.	$20.3 + 79.7 = 100\%$
Step 2: convert the percentages to masses in grams (assuming 100 g of the compound):	Mass of Al in 100 g = 20.3% $\rightarrow 20.3$ Mass of Cl in 100 g = 79.7% $\rightarrow 79.7$
Step 3: convert from the mass in grams to amount (in mol) by dividing each mass by the molar mass for that element:	$n(\text{Al}) = \frac{20.3}{26.98}$ $= 0.752 \text{ mol Al}$ $n(\text{Cl}) = \frac{79.7}{35.45}$ $= 2.25 \text{ mol Cl}$
Step 4: divide each value by the smallest amount in mol.	$\text{Al: } \frac{0.752}{0.752} = 1$ $\text{Cl: } \frac{2.25}{0.752} = 3$
Step 5: state the empirical formula.	This gives the lowest whole number ratio, so the empirical formula is: AlCl_3

Solution steps	Calculations
Step 6: find the empirical formula mass:	The empirical formula mass of AlO $(1 \times 26.98 \text{ g mol}^{-1}) + (3 \times 35.45 \text{ g mol}^{-1})$
Step 7: divide the relative formula mass by the empirical formula mass to see the whole number multiple	$\frac{267}{133.33} = 2.00$
Step 8: state the answer with appropriate units and the number of significant figures used in rounding:	So the molecular formula is twice hence the molecular formula is Al_2O_3

Study skills

Understand that the mass of the molecular formula must be a whole number multiple (including 1) of the empirical formula. If your molecular formula divided by the empirical formula is not a whole number (or very close to a whole number when rounded), then you have made an error. Possible errors could include using the incorrect masses from the periodic table, a calculation error or a rounding error. Your answer should always make sense for these types of calculations.

Concept

The empirical formula of a compound gives the simplest ratio of atoms (or ions) of each element present in that compound.

The molecular formula gives the actual number of atoms (or ions) of each element present in a molecule.

To determine the actual molecular formula from the empirical formula, we need to know the molar mass of the compound.

Watch **Video 1** to learn how to calculate empirical and molecular formulae.

Calculating Empirical and Molecular Formula: What is a ma...



Video 1. Calculating empirical and molecular formulae.

Activity

- **IB learner profile attribute:** Risk-takers
- **Approaches to learning:** Thinking skills — Providing a reasoned argument to support conclusions
- **Time required to complete activity:** 20—25 minutes
- **Activity type:** Pair activity

Identify the mystery substance!

White crystals were found spilled in the laboratory. A group of four students were working at one laboratory bench and they all denied spilling anything. Your school has access to a mass spectrometer and the mysterious substance was analysed. Can you and your partner work quickly to identify the compound so that it can be cleaned up and disposed of safely?

Data:

Analysis of the mysterious substance reveals that it has the following chemical composition.

Table 1. Chemical composition of the mysterious su

% carbon	
% hydrogen	

% oxygen	
-----------------	--

The students working in that area were using the following substances:

Table 2. Substances used by the students.

Citric acid	O
Sucrose	M

Prepare a report with your analysis of the mysterious substance with sufficient evidence to support your identification. Research the correct clean up and disposal method for this substance, according to your local waste disposal guidelines. Good luck!

Molar concentration of a solution

Learning outcomes

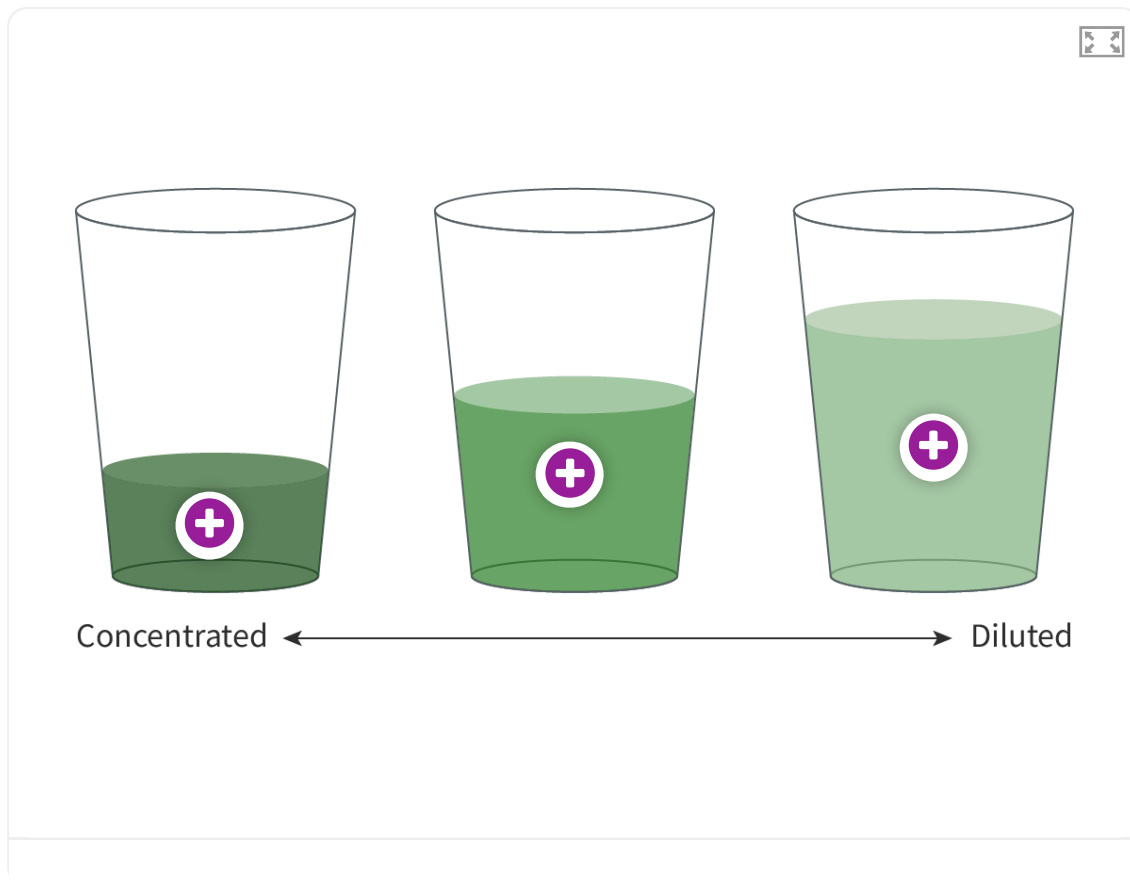
By the end of this section you should be able to:

- Describe solutions qualitatively as dilute and concentrated.
- Solve problems involving molar concentration, amount of solute and volume of solution.
- Identify laboratory equipment required to accurately prepare a solution of known concentration.

Recall from section S1.1.1a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>) that matter can be classified as a pure substance or a mixture. So far, you have learned how to calculate the number of elementary entities for pure substances, but what about mixtures? The most common mixture you will encounter in chemistry is a homogenous mixture made by dissolving a solute into a solvent, forming an aqueous solution. You have encountered aqueous solutions before in everyday life. Adding salt or sugar to water and stirring until you can no longer see the crystals is an example of an aqueous solution. Even the water you drink is not made up of only water molecules – there are many dissolved ions and even oxygen molecules dissolved in the water – you cannot see them, but they are there!

In a mixture, we express the quantity of the solute dissolved in the solution as a ratio, known as the concentration. You might be familiar with qualitative descriptions for describing the concentration of a solution. For example, if you have ever prepared a drink from a syrup or have made lemonade from lemon juice and sugar, you have taken a concentrated form of the drink that has a very strong flavour and added water to make a diluted form of the drink that has a weaker flavour. Concentrated and dilute are qualitative terms to describe concentration, but we require quantitative measurements so that we can relate concentration to the number of elementary entities. Can you think of some reasons why the precise concentration of a solution might be important?

Try **Interactive 1** to see how the concentration can be changed by the addition of solute and solvent.



Interactive 1. Dilute or Concentrated?

More information for interactive 1

An interactive illustration features three glasses containing solutions of different concentrations. The first glass is quarter-filled, the second glass is half-filled while the third glass is filled up to three-fourths with the solution. The first glass holds the most concentrated solution, indicated by its dark green color. The second glass contains a less concentrated solution than the first, indicated by its medium green tone. The third glass has a more diluted solution, indicated by its light green color.

There is a double-headed arrow at the bottom between the terms concentration and dilution, with concentration labeled on the left, pointing toward the first glass, and dilution labeled on the right, pointing toward the third glass.

Each glass has a plus icon, representing interactive hotspots. Users can click on these hotspots to reveal close-up views of the corresponding solutions. Clicking on the hotspot shows the close-up view of the corresponding glass, with the semantic zooming of the solute and solvent particles in that solution. The solvent particles are represented by small white circles while the solute particles are represented by small green circles. There is also text displayed beside the solvent and solute particles for each hotspot.

The text displayed for each hotspot is as follows:

Hotspot on first glass: Solvent particles, Solute Particles (17)

Hotspot on second glass: Solvent particles, Solute particles (16)

Hotspot on third glass: Solvent particles, Solute particles (15)

The number in parentheses beside the solute particles represents the number of solute

particles in that solution.

This interactive helps users understand the concept of solution concentration and how it varies with the amount of solute present.

Vinegar is a product that you might have in your house because it is a very versatile substance. It can be used in cooking to add flavour to sauces and salads, as a cleaning product for glass and ceramic tiles and can even be used to relieve the sting of an insect bite. Vinegar is an aqueous solution containing ethanoic acid, also known as acetic acid, a substance which you likely also have in your school laboratory. Laboratory grade ethanoic acid is very concentrated and must be handled with extreme caution, but household vinegar is diluted to the point that it is safe to ingest or make contact with the skin. The difference is the concentration (**Figure 1**).



Credit: Kittisak Kaewchalun, [Getty Images](#)

(<https://www.gettyimages.co.uk/detail/photo/acetic-acid-in-bottle-chemical-in-the-laboratory-royalty-free-image/1474288946>)



Credit: Eskay Lim / EyeEm, [Getty Images](#)

(<https://www.gettyimages.co.uk/detail/photo/close-up-of-various->

[vinegars-on-table-against-white-royalty-free-image/963037322\)](#)

Figure 1. Household vinegar is a diluted solution of acetic acid.

Examine the labels on a few different items in your house – medication, cleaning products and toiletries. Notice whether the quantities of the active ingredients are listed on the label. When concentration is listed on a label, it must be expressed using measurements that most consumers would understand, such as grams and litres, and is often expressed as a percentage concentration

$\left(\% \frac{\text{volume}}{\text{volume}} \text{ or } \% \frac{\text{weight}}{\text{volume}} \right)$. Recall that one litre is also expressed as a cubic decimetre, dm^3 .

Solutions are prepared by dissolving a solute in a solvent. We accurately measure the mass of the solute and the total volume of the solution, one way of expressing solution concentration is mass per volume. For example, if a salt solution contains 1.00 g of salt dissolved in 1.00 dm^3 of solution, the concentration would be expressed as 1.00 g dm^{-3} (**Figure 2**).

The figure shows the formula for concentration: $C = \frac{m}{V}$. Annotations with leader lines point to each part of the formula:

- Pointing to C : Concentration, expressed as g dm^{-3}
- Pointing to m : Mass of solute, measured in grams
- Pointing to V : Total volume of solution, measured in dm^3

Figure 2. Concentration is expressed as mass of solute per unit volume of solution.

 More information for figure 2

The image displays a formula for concentration represented as $C = m/v$, where 'C' denotes concentration, 'm' represents the mass of the solute, and 'v' indicates the volume of the solution. Around the formula, there are annotations explaining each component: "Concentration, expressed as g dm^{-3} ," pointing to 'C'; "Mass of solute, measured in grams," pointing to 'm'; and "Total volume of solution, measured in dm^3 ," pointing to 'v'. These annotations clarify that the concentration is expressed in grams per cubic decimeter, the mass is in grams, and the volume is in cubic decimeters.

[Generated by AI]

Try using the concentration formula yourself in these worked examples.

Worked example 1

Calculate the mass of sodium chloride, NaCl, required to make 2.00 dm³ of 0.800 g dm⁻³ solution.

Solution steps	Calculation
Step 1: first, identify the formula to use.	$C = \frac{m}{V}$
Step 2: rearrange the formula to make m the subject.	$m = C \times V$
Step 3: substitute in the known values.	$m = 0.800$
Step 4: state the answer with appropriate units and the number of significant figures used in rounding.	$= 1.60 \text{ g N}$
Step 5: state the answer with appropriate units and the number of significant figures used in rounding	Therefore, prepare thi

Worked example 2

Calculate the volume of solution that can be prepared when 300.0 mg of citric acid is diluted to a concentration of 0.500 g dm⁻³ solution.

Solution steps	Calculation
Step 1: first, identify the formula to use.	$C = \frac{m}{V}$

Solution steps	Calcula
Step 2: rearrange the formula to make V the subject.	$V = \frac{m}{C}$
Step 3: convert solute from mg to g.	$\frac{300}{1000} =$
Step 4: substitute in the known values.	$V = \frac{0.}{0.}$
Step 5: state the answer with appropriate units and the number of significant figures used in rounding.	$= 0.600$
Step 6: state the answer with appropriate units and the number of significant figures used in rounding.	Therefo can be

As a chemist, you must also be able to express concentration in terms of moles per unit volume, since we use moles as a way of grouping and counting elementary entities. Recall in [section S1.4.3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-mass-m-id-44262/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-mass-m-id-44262/>) that you learned how to calculate the amount in moles from a mass in grams. **Figure 3** has the formula for calculating the concentration of a solution in mol dm^{-3} .

The diagram shows the formula $C = \frac{n}{V}$ with three annotations:

- An arrow points from the text "Concentration, expressed as mol dm^{-3} " to the variable C .
- An arrow points from the text "Moles of solute (mol)" to the variable n .
- An arrow points from the text "Total volume of solution, measured in dm^3 " to the variable V .

Figure 3. Concentration can also be expressed in moles per unit volume.

 More information for figure 3

The image is a diagram showing the formula for calculating concentration in terms of moles per unit volume. It displays the mathematical expression: $C = n / V$. Three annotations are present:

1. 'C' is labeled as Concentration, expressed as mol dm^{-3} .
2. 'n' denotes the Moles of solute (mol).
3. 'V' represents the Total volume of solution, measured in dm^3 .

This formula indicates that concentration (C) is calculated by dividing the number of moles of solute (n) by the volume of the solution (V) in cubic decimeters (dm^3).

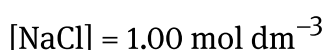
[Generated by AI]

Worked example 3

Calculate the molar concentration, in mol dm^{-3} , of the 0.500 g dm^{-3} citric acid, $\text{C}_6\text{H}_8\text{O}_7$ solution.

Solution steps	Calculations
Step 1: first, calculate the molar mass of citric acid.	$M = (6 \times 12.01) + (8 \times 1.01) + (7 \times 16.00)$ $= 192.14 \text{ g mol}^{-1}$
Step 2: calculate the amount (in mol) of citric acid	$\frac{0.500}{192.14} = 0.00260 \text{ mol}$
Step 3: divide the amount (in mol) by the volume (in dm^3)	$\frac{0.00260}{1} = 0.00260 \text{ mol dm}^{-3}$
Step 4: state the answer with appropriate units and the number of significant figures used in rounding.	The molar concentration is $0.00260 \text{ mol dm}^{-3}$.

The use of square brackets, [], is often used to express molar concentration, also known as molarity. The formula of the solute is placed in the square brackets. For example, a 1.00 mol dm^{-3} solution of sodium chloride, NaCl, can be expressed as:



Worked example 4

Calculate the molar concentration of a solution that is prepared by dissolving 5.00 g of sodium chloride, NaCl, in 2.00 dm³ of water.

Solution steps	Calculations
Step 1: first, convert the mass of sodium chloride into amount (in mol) using the molar mass.	$n = \frac{5.00}{58.45}$ $= 0.0855 \text{ mol NaCl}$
Step 2: identify the formula to be used.	$n = CV$
Step 3: rearrange to make C the subject.	$C = \frac{n}{V}$
Step 4: substitute the known values.	$C = \frac{0.0855}{2.00}$ $= 0.0428 \text{ mol dm}^{-3}$
Step 5: state the answer with appropriate units and the number of significant figures used in rounding.	Therefore, the concentration is 0.0428 mol dm ⁻³ , or it can be written as [NaCl] = 0.0428 mol dm ⁻³ .

Study skills

The formula $n = CV$ is included in the DP chemistry data booklet, so you do not have to memorise it. When concentration is expressed using moles, it is referred to as molar concentration or molarity. The units are expressed as mol dm⁻³, mol L⁻¹ or simply M, where M represents molar concentration, or molarity. For example, a 1.00 M solution is said to have a concentration of 'one molar'. Your exam paper questions will always show units of mol dm⁻³, but you may show your work or display an answer containing units of mol/L or M on your exam paper or on your laboratory work.

Worked example 5

Determine the mass of magnesium bromide, MgBr_2 , required to make 200.0 cm^3 of $0.0500 \text{ mol dm}^{-3}$ solution.

This calculation requires two steps – first to determine the amount of the solute, in mol, required for the solution, then converting that to an amount in grams.

Solution steps	Calculations
Step 1: identify the formula to be used.	$n = CV$
Step 2: change the units of volume from cm^3 to dm^3 .	$\frac{200}{1000} = 0.200 \text{ dm}^3$
Step 3: substitute the known values.	$n = 0.0500 \times 0.2000$ $= 0.0100 \text{ mol MgBr}_2$ The amount of MgBr_2 in solution is 0.0100 mol, so in grams.
Step 5: calculate the molar mass.	$M(\text{MgBr}_2) = (1 \times 24.3) + (2 \times 79.9)$ $= 184.1$
Step 6: calculate the mass of the compound.	$\text{mass} = 0.0100 \times 184.1$ $= 1.84 \text{ g}$
Step 7: state the answer with appropriate units and the number of significant figures used in rounding.	The mass of MgBr_2 required for the solution is 1.84 g .

Practical skills

Tool 1: Experimental techniques — Measuring variables

Preparing a standard solution of a specific concentration is a skill that requires careful measuring and special equipment.

Preparing a standard solution

When you are sick and require medication, you want to be certain that you are taking the correct dose – if the quantity is too low, it will not be effective in treating your illness and if it is too high, the medication can cause toxic effects in the body. You might remember as a small child that oral medication is often given in a liquid form, rather than in pill form, because it is easier for children to swallow. The doctor calculates the dosage required and the pharmacist prepares a suspension of the active medicinal ingredient in a syrup. If either the doctor or the pharmacist makes an error, the medicine will not be effective in treating the illness. Likewise, preparing solutions for chemical reactions requires careful measuring to be sure the concentration is correct.

The mass of the solute must be measured accurately on a mass balance – to the nearest 0.01 or 0.001 g. It is then combined with a small amount of distilled water before being transferred to a volumetric flask. A volumetric flask is a specialised piece of glassware that measures precisely to one volume measurement only (**Figure 4**).

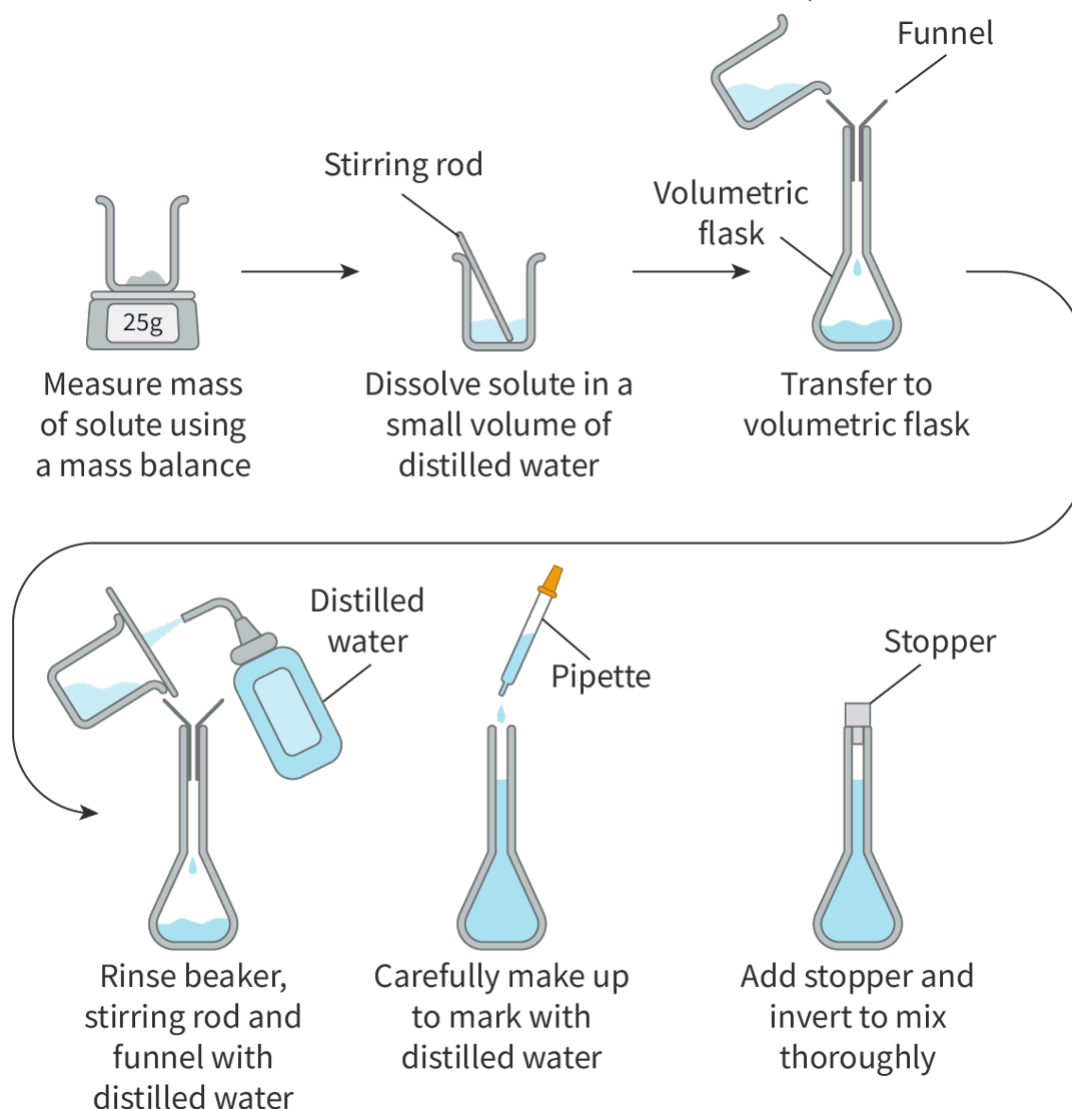


Figure 4. Preparation of a standard solution requires careful measuring to ensure an accurate concentration.

 More information for figure 4

The flowchart illustrates the process of preparing a standard solution in several steps. It begins with measuring the mass of the solute using a mass balance, indicating a value of 25g. Next, the solute is dissolved in a small volume of distilled water using a stirring rod. This solution is then transferred to a volumetric flask through a funnel.

After transferring, the procedure involves rinsing the beaker, stirring rod, and funnel with distilled water to ensure all solute is transferred. Then, using a pipette, distilled water is carefully added to make up to the mark on the volumetric flask. Finally, a stopper is added, and the flask is inverted to mix the solution thoroughly, completing the preparation process.

[Generated by AI]

Practical skills

Tool 1: Experimental techniques — Measuring variables

Another method of preparing a solution involves making a dilution using a solution that is already prepared. This is sometimes referred to as making a 'secondary' solution using a more concentrated 'primary' solution. Since the dilution is made by adding more solvent and the amount of solute does not change, the amount in mol remains the same for both the primary and secondary solutions.

Using $n = C_1V_1$ for the primary solution and $n = C_2V_2$ for the secondary solution, we can combine these two equations into one, setting the number of moles equal.

$$C_1V_1 = n = C_2V_2, \text{ therefore } C_1V_1 = C_2V_2$$

For example, 1.00 dm^3 of a $0.300 \text{ mol dm}^{-3}$ solution of NaOH was diluted to a volume of 2.00 dm^3 . Determine the concentration of the diluted solution.

$$\begin{aligned} C_1V_1 &= C_2V_2 \\ 0.300 \times 1.00 &= C_2 \times 2.00 \\ C_2 &= 0.150 \text{ mol dm}^{-3} \end{aligned}$$

We can see that the concentration of the secondary solution is less than the primary solution, which makes sense because it has been diluted.

Creativity, activity, service

Strand: Service

Learning outcomes:

- Demonstrate how to initiate and plan a CAS experience.
- Demonstrate engagement with issues of global significance.

Access to clean water is number 6 (<https://www.un.org/sustainabledevelopment/water-and-sanitation/>) of the United Nations' Sustainable Development Goals (SDGs). According to the UN, more than 800,000 people die every year from a lack of access to clean drinking water and inadequate sanitation. Reasons for this include climate change, water pollution and mismanagement of available water supplies. Contaminants (<https://www.epa.gov/ccl/types-drinking-water-contaminants>)

found in polluted water can include physical contaminants such as rubbish (trash), chemical contaminants such as pesticides and biological contaminants such as harmful microorganisms.

As an IB student and a global citizen, what can you do to help those people affected by a lack of access to clean drinking water? One possible CAS experience could be to raise awareness of the situation through a poster campaign at your school or in your local community. What other ways can you think of to help achieve these goals?

Use the activity in the next section to investigate molarity.

Avogadro's law of combining volumes

Learning outcomes

By the end of this section you should be able to:

- Describe the factors that affect the number of particles of gas in a sealed container.
- State Avogadro's law that there are equal volumes of all gases at constant temperature and pressure conditions.
- Solve problems using the mole ratio for gaseous reactants and products in reaction equations.

So far, you have learned how to calculate the number of elementary entities for solids and aqueous solutions. What about for gases? Recall in [section S1.1.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>) that gas particles are very far apart from each other and that the container they are held in is mostly empty space. How can we determine the number of elementary entities that are inside? As these particles are constantly in motion, it would be very difficult to count them, even if we could see them. We must make several measurements that affect the number of particles that can fit inside the container while still allowing the substance inside to behave as a gas.

Factors affecting gases

When a sample of gas is sealed inside a closed container, the particles are constantly in motion because they have kinetic energy. If one particle hits another particle or the sides of the container, it bounces off in another direction with no change to the amount of energy it has. The number of gas particles that can fit inside a closed container is affected by a few different variables:

- The volume of the container – the larger the container, the more particles can fit.
- The temperature of the gas – greater temperatures mean more kinetic energy and therefore greater speeds the particles will travel. At low temperatures, the particles move slowly, so more

particles can fit inside the container. At high temperatures, the particles need more space to move, therefore fewer particles can be accommodated.

- The pressure of the gas. Pressure is a measurement of the frequency of the collisions of the gas particles with the sides of the container. At low pressures, there are fewer collisions with the sides of the container because there are fewer particles present. When more particles are present, the pressure will be higher.

International Mindedness

Pressure and temperature affect the volume a gas will occupy. These two conditions can vary greatly throughout the day in a single location and drastically in different cities around the world. When scientists conduct experiments involving gases, temperature and pressure conditions are carefully controlled and monitored so that these experiments can be reproduced with consistent results. There are also agreed upon standard temperature and pressure conditions that scientists use for consistency with experimental conditions. You will learn more about these in section S1.5.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-volume-of-an-ideal-gas-and-gas-laws-id-44352/>).

These three variables together determine the number of gaseous elementary particles that can fit in a container. Since the container is mostly empty space and the gas particles have negligible volumes, an equal number of gas particles will occupy the same volume measured under the same conditions of temperature and pressure. This is known as Avogadro's law.

In **Figure 1**, the balloons all contain different gases at the same temperature and pressure and occupy the same volume. What do you notice about the number of particles inside?

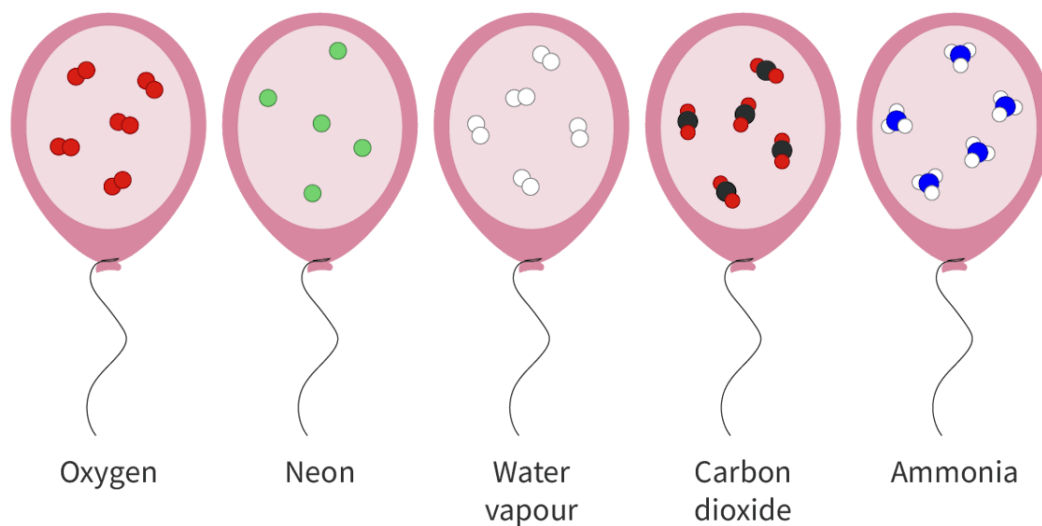


Figure 1. The balloons contain different gases but all have the same volume and contain the same amount of particles.

There are the same number of gas particles at constant temperature, pressure and volume. Do all of these balloons have the same mass? What would affect the mass that each of these balloons have?

No, these balloons do not have the same mass, since different atoms have different masses. We must use the molar mass and the amount, in mol, to determine the mass of each balloon.

Molar masses for different gases.

Balloon	Gas	Molar mass
A	O ₂	32
B	Ne	20
C	H ₂ O	18
D	CO ₂	44
E	NH ₃	17

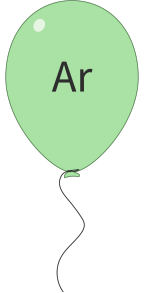
Therefore, balloon D would have the greatest mass and balloon E would have the lowest mass.

Arrange these balloons **Interactive 1** from smallest to greatest mass.

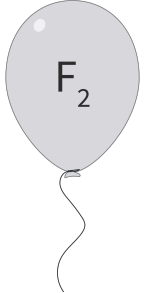
Smallest mass

Mid mass

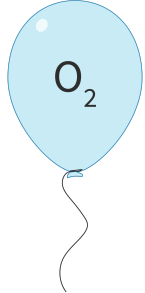
Largest mass



Ar



F₂

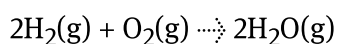


O₂

☒ Check

Interactive 1. Ordering Gases by Mass.

Avogadro's law is helpful for determining the quantity of gases involved in chemical reactions. For example, water vapour can be made from hydrogen and oxygen gases according to the reaction below:



Using Avogadro's law, we can assume that under constant temperature and pressure conditions, twice the volume of hydrogen gas is required to react with oxygen gas and that the quantity of water vapour produced will be equal to the volume of hydrogen gas that reacted and twice the volume of oxygen gas that reacted. For example, if 1.00 dm³ of hydrogen gas and 0.50 dm³ of oxygen gas reacted, then 1.00 dm³ of water vapour will be produced (**Figure 2**).

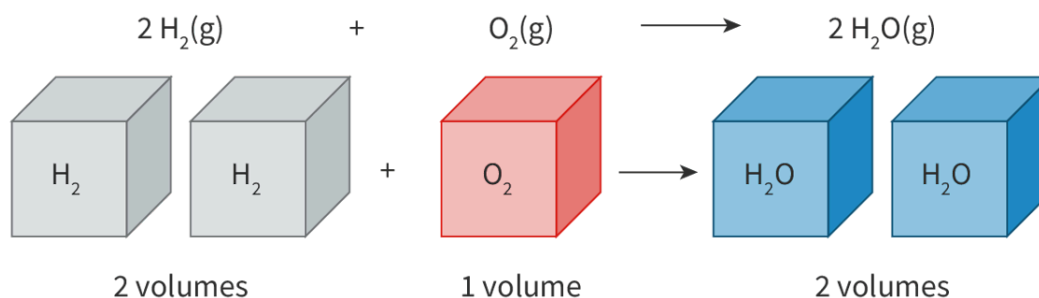


Figure 2. The molar volume of gases involved in a chemical reaction can be predicted using Avogadro's law.

 More information for figure 2

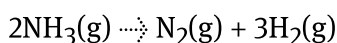
The diagram depicts Avogadro's law using a visual representation of gas volumes in a chemical reaction. It shows two gray cubes labeled H_2 (representing hydrogen gas) with the text "2 volumes" beneath them, indicating two volumes of hydrogen gas. Next to them is a red cube labeled O_2 (for oxygen gas) with "1 volume" beneath it, representing one volume of oxygen gas. An arrow points rightward from the hydrogen and oxygen cubes, leading to two blue cubes labeled H_2O (representing water vapor) with the text "2 volumes" beneath, indicating the formation of two volumes of water vapor. The layout visually supports the explanation that twice the volume of hydrogen gas combines with one volume of oxygen gas to produce a volume of water vapor equal to the hydrogen gas's volume, adhering to Avogadro's law.

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This law will become very helpful to you when you learn more about chemical reactions in [section R2.1.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chemical-equations-id-43480/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chemical-equations-id-43480/).

Worked example 1

Determine the volumes of hydrogen and nitrogen gases produced when 5.00 dm^3 of gaseous ammonia decomposes under constant temperature and pressure according to the reaction equation below:



Solution steps	Calculations
Step 1: use Avogadro's law for N_2 and NH_3 .	1 volume of $\text{N}_2(\text{g})$ is pro of $\text{NH}_3(\text{g})$ that decomp $2 \text{ NH}_3(\text{g}) : 1 \text{ N}_2(\text{g})$ $5.00 \text{ dm}^3 \text{ NH}_3(\text{g}) :$
Step 2: use Avogadro's law for H_2 and NH_3 .	3 volume of $\text{H}_2(\text{g})$ is pro of $\text{NH}_3(\text{g})$ that decomp $2 \text{ NH}_3(\text{g}) : 3 \text{ H}_2(\text{g})$ $5.00 \text{ dm}^3 \text{ NH}_3(\text{g}) :$
Step 3: state the answer with appropriate units and the number of significant figures used in rounding.	So $2.50 \text{ dm}^3 \text{ N}_2(\text{g})$ and when $5.00 \text{ dm}^3 \text{ NH}_3(\text{g})$

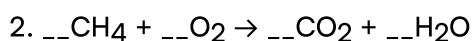
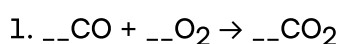
Study skills

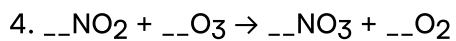
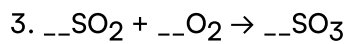
Avogadro's law only applies to gaseous substances in a reaction equation. Be sure to check that you are only applying the ratio to species in the gaseous state and not to substances that are solids, liquids or aqueous solutions.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 15–20 minutes
- **Activity type:** Individual activity

Balance the reaction equations and model the chemical reactions using cubes to model 1 dm^3 units.

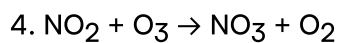
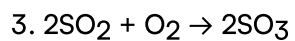
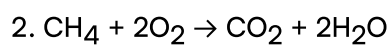
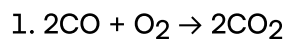




Using your answers above, complete the chart below with the quantities of each gas involved in the chemical reaction:

CO

O₂CO₂1.00 dm³CH₄O₂CO₂H₂O2.00 dm³SO₂O₂SO₃2.50 dm³NO₂O₃NO₃O₂

10.0 dm³

Using your answers above, complete the chart below with the quantities of each gas involved in the chemical reaction:

CO

O₂CO₂1.00 dm³0.500 dm³1.00 dm³CH₄O₂CO₂H₂O1.00 dm³2.00 dm³1.00 dm³2.00 dm³SO₂O₂SO₃

2.50 dm^3 1.25 dm^3 2.50 dm^3 NO_2 O_3 NO_3 O_2 10.0 dm^3 10.0 dm^3 10.0 dm^3 10.0 dm^3